

EDUCATION

- 2024.09 – Present** **University of Nottingham** Ph.D. Candidate in Computer Science
Strategic Scholarship (Full Scholarship)
Research Interests: Quantum Key Distribution, Post-Quantum Cryptography, Lattice-Based Digital Signature Schemes, 5G Authentication Protocols, 5G Secure Communication, and AI Privacy Preservation.
- 2020.09 – 2024.06** **Northeastern University (China)** B.Eng. in Communication Engineering
Overall GPA: 86.93/100 (Rank: 9/118).

PUBLICATIONS

- 2026 JCR Q1** **A Survey on the Security of 5G and Beyond Networks in the Post-Quantum Era** (Ximing Zhang, Pushpendu Kar, Chiew Foong Kwong, David Chieng, Alejandro Guerra-Manzanares.) *Computer Networks* (Accepted for publication.)
Key Contributions:
 - We systematically review the literature in this specific field. We analyzed evidence from the reviewed literature to address three research questions, resulting in a novel taxonomy to categorize publications in this area and a vision for a quantum-safe 6G network. Moreover, we discussed several open challenges in 5G, PQC, and QKD, along with future work.
- 2026 JCR Q2** **A Survey on Quantum Key Distribution: Foundations and Future Directions** (Ximing Zhang, Pushpendu Kar, Chiew Foong Kwong, David Chieng, Alejandro Guerra-Manzanares.) *Computer Communications*.
Key Contributions:
 - We provide a mathematical foundation for quantum key distribution (QKD), and review the state-of-the-art research development in QKD, including popular QKD protocols and their security proofs, along with cutting-edge fiber-based, free-space, and chip-based QKD advances.
- 2026** **Securing 5G User Plane Using Post-Quantum DTLS Protocol** (Ximing Zhang, Pushpendu Kar, Chiew Foong Kwong, David Chieng, Theodoros A. Tsiftsis, Alejandro Guerra-Manzanares.) *2026 IEEE Symposium on Computers and Communications (ISCC), Algarve, Portugal*.
Key Contributions:
 - We propose a secure transmission scheme for the 5G user-plane N3 interface based on PQ-DTLS 1.3. PQ-KEM and PQ-DSA provide post-quantum security, while mutual authentication and certificate verification ensure authenticity and resistance to man-in-the-middle and replay attacks.
- 2023** **Intelligent Network Threat Detection Engine Based on the Open-Source GPT-2 Model.** (Ximing Zhang, Tong Chen, Jingyi Wu, Qilong Yu.) *International Conference on Computer Science and Automation Technology (CSAT)*, Shanghai, China.

PATENT

- Feb. 10, 2026** **Post-Quantum Secure Transmission Method, System, Device, and Medium for the 5G User Plane** (Patent No.: CN121509995A). **Granted**
Inventors: Ximing Zhang; Pushpendu Kar; Chiew Foong Kwong; Hengzhi Zhan; Alejandro Guerra-Manzanares
Description: This invention introduces a quantum-resistant secure channel in the 5G User Plane incorporating post-quantum key agreement, quantum-resistant encryption, integrity verification, and identity authentication.

RESEARCH EXPERIENCE

- 2026.01 – Present** **Privacy-Preserving 5G Authentication Protocol**
Background: Current 5G authentication protocols rely on ECC and lack post-quantum security and privacy protection.
Responsibilities: Develop new authentication and privacy-preserving protocols; use symbolic verification tools such as Tamarin to support protocol security proofs.
- 2026.01 – Present** **Natively-Secure Quantum-Safe 5G User Plane Protocol**
Background: The 5G user plane protocol lacks quantum resistance and native security considerations.
Responsibilities: Design natively secure and quantum-safe 5G user plane protocols using post-quantum cryptography and quantum key distribution.
- 2024.09 – 2025.12** **Post-Quantum Secure Transmission Protocol for the 5G User Plane**
Background: The GPRS Tunneling Protocol currently used on the user plane lacks confidentiality and integrity protection.
Responsibilities: Used C/C++-based open-source 5G frameworks, including Open5GS and UERANSIM, for simulation and secondary development to integrate and benchmark a new secure user-plane transmission protocol.
Outcome: Published in the IEEE ISCC 2026 conference proceedings as listed above. The protocol preserved user-plane throughput and end-to-end latency while adding only **2.5 ms** of initial authentication and key-agreement latency and **1.32 ms** of CPU processing latency. It provides **replay-attack resistance, post-quantum secrecy, confidentiality, and authenticity**, together with a **security probability bound**.
- 2025.12– 2026.03** **Blockchain- and IoT-Integrated Waste Management System (Funded by: China International Talent Exchange Foundation.)** **Role:** Key Participant (Principal Investigator: Dr. Pushpendu Kar).
Background: This project aims to develop a sustainable waste management system by integrating IoT sensors with blockchain. The system enables real-time monitoring of waste generation and collection, supporting efficient waste management and environmental sustainability.
Responsibilities: To help undergraduate and master's students develop research and programming skills in IoT and blockchain, delivered a lecture on "5G IoT (NB-IoT)" and led hands-on experiments.
- 2022.09 – 2023.09** **RISC-V-Based Network Threat Detection Engine (Funded by: Student Innovation and Entrepreneurship Center, Northeastern University at Qinhuangdao)**
Background: Investigated how to build a distributed and efficient network threat detection engine for IoT devices based on the RISC-V architecture.
Responsibilities and Outcomes: I published a conference paper. I built a Suricata traffic detection engine on a QEMU-emulated RISC-V Ubuntu virtual machine, designed custom detection rules, processed data, and implemented front-end visualization. I periodically collected and cleaned data to fine-tune an open-source GPT-2 model, then applied the fine-tuned model to real-time traffic detection, achieving over 90%.

OUTSIDE STUDY ACTIVITIES

- 2026 08.08-08.11** **Quantum Artificial Intelligence Summer School, Changsha, China**
I have been selected as one of the 100 students from around China to participate. During the summer school, I learned classical quantum algorithms, quantum error correction, and cutting-edge developments in quantum artificial intelligence.

TEACHING EXPERIENCE

- 2026.02 – 2026.06** **Graduate Teaching Assistant, Mobile Device Computing (COMP3059)**
Assisted with laboratory sessions and supported students in developing Android/Java applications.
- 2025.09 – 2025.12** **Graduate Teaching Assistant, Programming and Algorithms (COMP1038)**

Assisted with laboratory sessions and supported students in C programming, data structures, and algorithms.

2022.02 – 2025.05 **Part-Time Lecturer, HD Edu.**
Taught core courses including Deep Learning, Number Theory and Cryptography, Post-Quantum Cryptography, Cybersecurity, Penetration Test, Analog and Digital Communications, Digital Signal Processing, Partial Differential Equations, Complex Analysis, Computer Architecture, and Computer Networks.

INTERNSHIP EXPERIENCE

2024.03 – 2024.06 **Computer Vision Algorithm Engineer**, Nanjing Dingzi Network Technology Co., Ltd.
Responsibilities: Selected suitable models for project requirements, including classification, detection, and segmentation models; designed data-cleaning and annotation workflows; trained, optimized, and evaluated models; deployed and accelerated models; used models to label images and evaluated overall output quality.

2022.03 – 2022.08 **Cybersecurity Engineer**, NSFOCUS Technologies Group Co., Ltd.
Responsibilities: Co-authored the vocational skills textbook *Cybersecurity Services 1+X*, covering cybersecurity fundamentals, penetration-testing fundamentals, Metasploit, web security, firewall principles and practice, and log auditing.

2021.08 – 2021.11 **Engineering Intern**, Beijing Kedinuoyi Technology Co., Ltd.
Responsibilities: Conducted IoT security testing and maintained servers to ensure secure operation.

HONORS AND AWARDS

2024.05 Full Ph.D. Scholarship
2023.05 University Second-Class Comprehensive Scholarship
2023.04 Innovation and Entrepreneurship Scholarship, Northeastern University at Qinhuangdao
2022.08 Third Prize, North China Regional Contest, 2022 Network Technology Challenge, China Collegiate Computing Contest
2022.05&2021.10&2021.05 University Third-Class Comprehensive Scholarship
2021.10 14th Place, Conflux Summer Blockchain Boot Camp

TECHNICAL SKILLS

- **Programming Languages:** Proficient in C/C++, Java (Android), and Python libraries including Cryptography, NumPy, and PyTorch; familiar with Rust, Solidity, PHP, and Verilog.
- **Tools and Frameworks:** Proficient in PyTorch, SageMath, UERANSIM, Open5GS, TikZ, OpenSSL, and wolfSSL; familiar with Vitis and Vivado.
- **English:** IELTS 7.0 (Listening 7.0, Reading 8.0, Writing 6.0, Speaking 6.0).
- **5G Standards:** Familiar with 3GPP TS 23.501, 23.501, 23.401, 33.501, 29.244, and 33.210.

ACTIVITIES AND INTERESTS

2021.10 – 2023.09 **CTF Player** Participated in CTF competitions, with a focus on cryptography and web security.
Interests Long-distance running, Piano, Calligraphy, and Badminton